Table 1: Summary Results of Simulation Study of Naive VS RRC-Corrected Effect Estimates of Cumulative Total Exposure

Generation of C	**Relative bias (%)**[a]		**Average estimated SD**[b]		**Monte-Carlo SD**[c]		**95% CI coverage**	
	Naive	RRC	Naive	RRC	Naive	RRC	Naive	RRC
Cox+Additive	Cox true $\beta = \log(1.5) \approx 0.405$, additive independent measurement error							
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	−50.49	−0.97	0.035	0.068	0.036	0.071	0.00	93.35
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	−49.89	0.35	0.035	0.070	0.034	0.068	0.00	95.50
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	−49.98	0.10	0.035	0.071	0.035	0.071	0.00	94.90
Cox+Multiplicative	Cox true $\beta = \log(1.5) \approx 0.405$, multiplicative independent measurement error							
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	−46.16	6.93	0.040	0.080	0.047	0.091	0.95	89.70
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	−47.64	4.83	0.038	0.076	0.038	0.076	0.05	93.65
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	−49.56	0.92	0.038	0.076	0.037	0.075	0.05	94.85
ERR+Additive	ERR true $\beta \approx 0.423$, additive independent measurement error							
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)[d]	−47.59	4.58	0.100	0.199	0.098	0.194	42.82	94.23
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)[d]	−46.93	6.16	0.101	0.203	0.101	0.201	43.72	95.23
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)[d]	−47.84	4.21	0.102	0.203	0.099	0.197	43.07	94.63
ERR+Multiplicative	ERR true $\beta \approx 0.423$, multiplicative independent measurement error							
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	−51.74	−3.02	0.052	0.107	0.055	0.110	6.45	92.20
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	−53.17	−5.48	0.054	0.111	0.054	0.111	6.70	90.85
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	−51.95	−3.35	0.050	0.102	0.051	0.103	5.25	92.20

[a] $100 \times \{\text{mean}(\hat{\beta}) - \beta\}/\beta$.
[b] Mean of the estimated standard errors: model-based for Naive, sandwich estimator for RRC.
[c] Empirical standard deviation of $\hat{\beta}$ across replications.
[d] Excludes replications in which a fit did not converge or ended on the boundary of the parameter space: 43, 8, 8 of 2,000 in the rows marked, in order.
Main study sample size n_1=6000, validation study sample size n_2=2000, study duration T^*=55, simulation repetition number B=2000, cumulative incidence rate about 10%. Parameter values are in Tables 2 and 3.